

Choosing a splicing score for a fixed minigene budget: an exploratory comparison of a supervised baseline, SpliceAI and Pangolin on MFASS

Study repository `rewire-bio/splicing-model-selection`

Author list to be confirmed by the study owner. Drafted with AI assistance; see Section 8.

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Status of this manuscript. This is a manuscript reporting *existing* computational results. Every number comes from a companion run executed on 29–30 September 2026, from the published annotation-matched study whose saved scores that run replays, or from the cited literature. Numbers were extracted and formatted from the archived outputs; nothing was recomputed for this manuscript. A fresh, independent reproduction is **pending**: the reproduction protocol in the repository (`protocol.md`, draft v2, unapproved) incorporates corrections from an automated AI methods review of draft v1 and has not been executed. No result here carries a verified or approved status. All intervals are exploratory and unadjusted, because the held-out outcomes had been inspected in earlier work.

Abstract

A laboratory that can run a fixed number k of minigene splicing assays needs a score that places as many splice-disrupting variants as possible in its first k . We compared a supervised gradient-boosted k -mer baseline trained on assay labels with two zero-shot splice specialists, SpliceAI and Pangolin, each with masking off and on, on the held-out arm of the MFASS exon-recognition assay. SpliceAI and Pangolin scores were not regenerated: they are a saved-score replay of published per-variant predictions from an annotation-matched study (GRCh38, GENCODE 44 canonical transcripts). On the 8,297 held-out variants that every configuration scored (314 disrupting, 460 exon/gene groups), the historical baseline `kmer_cons` placed 61 disrupting variants in its first 100, and the four specialist configurations placed 63 to 66. No difference at $k = 100$ was established: every paired 95% whole-group bootstrap interval for the precision difference includes zero (for unmasked Pangolin, $+0.040 [-0.038, +0.120]$), and together the intervals allow about 7 fewer to 13 more disruptions per 100. This is not equivalence. Against a validation-selected version of the baseline the point differences shrink to between $+0.000$ and $+0.030$. Pangolin had higher whole-list average precision ($+0.102 [+0.061, +0.142]$ for the unmasked configuration) and AUROC, and at a budget of 300, chosen after the budget curve had been seen, both Pangolin intervals exclude zero. That is a lead to test, not a finding. We also document a failure mode: 23 held-out variants have recorded alleles that are the complement of GRCh38, consistent with an inverted hg19-to-hg38 mapping that is not author-confirmed. The endpoint is exon recognition in a synthetic reporter, not patient splicing or pathogenicity.

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1 Introduction

Massively parallel minigene assays let a laboratory measure the splicing effect of many single-nucleotide variants (SNVs), but each construct still has a cost. A group with capacity for k constructs, and a candidate list much longer than k , has to rank the list. The quantity that matters for that decision is not a calibrated probability or a classification threshold but the number of splice-disrupting variants among the first k ranked candidates: the precision at budget k (P@ k).

Two kinds of score are available. A supervised model trained on labels from the same assay design can learn the reporter’s idiosyncrasies, but needs thousands of labels. Zero-shot splice specialists such as SpliceAI [1] and Pangolin [2] need no assay labels, but read the genome around the variant rather than the construct, and their outputs depend on annotation and masking settings. Earlier benchmarks have shown that the optimal score threshold for these tools varies between exons, regions and variant classes [3], which is one reason to rank at a budget rather than threshold a score.

This manuscript reports a comparison of these workflows on the MFASS assay [4] at fixed budgets. Its contributions are: (i) a P@ k comparison on one common scored population, with explicit tie ranges and paired whole-group bootstrap intervals, against both a historical and a validation-selected supervised baseline; (ii) a description of how the answer depends on the budget, with the post hoc budgets marked; (iii) a documented assembly-orientation failure mode: variants a pipeline would have scored without warning if allele checks were skipped; and (iv) a runnable companion that, in the original run, regenerated every control score and metric from pinned public inputs (independent reproduction is pending).

Two scope statements apply throughout. First, the comparison is between *workflows*, not architectures: inputs, training data, ensemble size, masking rules and score definitions all differ between the configurations. Second, the held-out outcomes had been inspected in earlier work on this benchmark, so every interval is exploratory and none is adjusted for multiple comparisons.

2 Background and related work

2.1 The MFASS assay

MFASS [4] places each variant in a 170 bp construct — a natural exon under 100 bp with at least 40 bp of upstream and 30 bp of downstream intron — inside a three-exon minigene reporter integrated once per HEK293T cell. Sorting cells by fluorescence gives an exon inclusion index. A splice-disrupting variant (SDV) lowers the index by at least 0.5 in an exon whose wild-type index is at least 0.5. Of 27,733 ExAC variants, 1,050 (3.8%) were disrupting, 83% of them outside canonical splice sites [4]. Figure 1 reproduces a published schematic of the reporter.

2.2 Splice specialists, masking and annotation

SpliceAI [1, 6] and Pangolin [2, 7] predict splice-site usage from sequence and report variant scores relative to annotated genes. Both offer a “mask” option, but equal flags are not equal operations: SpliceAI masks against the nearest annotated boundary and Pangolin against every annotated site in the window. The defaults also differ: SpliceAI’s `-M` is 0, Pangolin’s `-m` is True. A result reported for “SpliceAI” or “Pangolin” without these settings does not identify what was run. Smith and Kitzman [3] showed that masking and annotation choices change the set of variants each tool calls (Figure 2).

2.3 Thresholds versus budgets

Ranking at a budget avoids choosing a score threshold. In Smith and Kitzman’s benchmark of eight predictors on massively parallel splicing assays [3], the best cut-off changed between exons, regions and variant classes, and for SpliceAI, Pangolin and ConSpliceML was usually below the

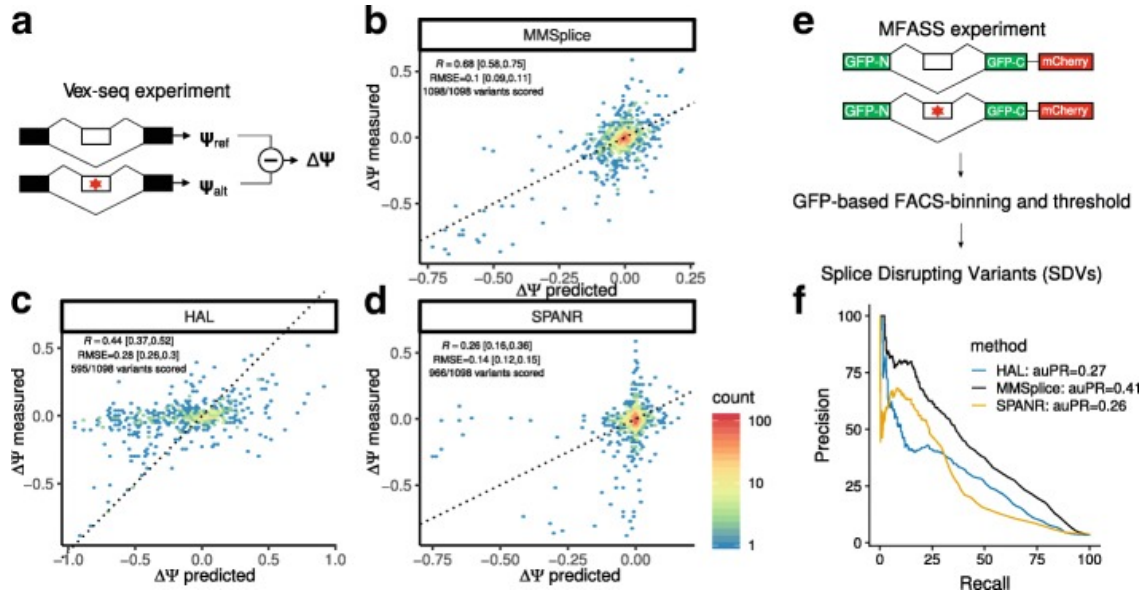


Figure 1: The MFASS reporter and earlier predictors on it. Panel e shows the MFASS design: a split-GFP minigene, sorting of cells into fluorescence bins, and the disruption threshold. Panels a–d show a different assay (Vex-seq). In panel f each tool’s precision–recall curve covers only the variants that tool could score, so the subsets differ; SpliceAI and Pangolin are absent. Source: Cheng et al., *Genome Biology* 2019, Fig. 2 [5], <https://pmc.ncbi.nlm.nih.gov/articles/PMC6396468/figure/Fig2/>, CC BY 4.0, unmodified.

developers’ recommended value (0.2 for SpliceAI and Pangolin in that benchmark; Figure 3). The SpliceAI cut-offs of ≥ 0.2 and ≤ 0.1 are ClinGen calibrations for clinical evidence [8], not rules for filling an assay budget; 0.2 is also the SpliceAI authors’ high-recall cut-off, and their README calls 0.5 “recommended” [6].

2.4 Other benchmarks

MMSplice [5] reported precision–recall curves on MFASS for MMSplice, HAL and SPANR, each tool on the subset of variants it could score (Figure 1, panel f); SpliceAI and Pangolin are absent from that comparison. The CAGI5 splicing assessors [9] caution that predicting a splicing assay “cannot directly imply clinical significance”, a limit that applies equally here (Section 6).

3 Materials and methods

3.1 Data and split

The cohort was built by the companion’s build step from the MFASS tables at the authors’ repository, commit 9a8e4f2 [10]. The build reads 32,669 source rows and keeps 27,733 eligible variants (1,050 disrupting), matching the published counts; the paper’s 2,198 exons are counted before the eligibility filter, and the eligible variants span 2,185. Every eligible row passes an assay-pair check (reference and mutant constructs differ only at the recorded position, with alleles consistent with strand). The legacy `sequence` column is reverse-complemented in 7,770 rows, a detail that once led an earlier baseline to centre its k -mer window on the wrong base [11].

The split is the benchmark’s canonical `split-v2` [11]: whole exon-and-gene groups go to one arm, and no Ensembl gene identifier crosses arms. The training arm has 19,409 variants (735 disrupting) in 1,127 groups; the held-out arm has 8,324 variants (315 disrupting) in 463 groups. These build counts are reported as article text from the original run; the build log is not part of the archived outputs.

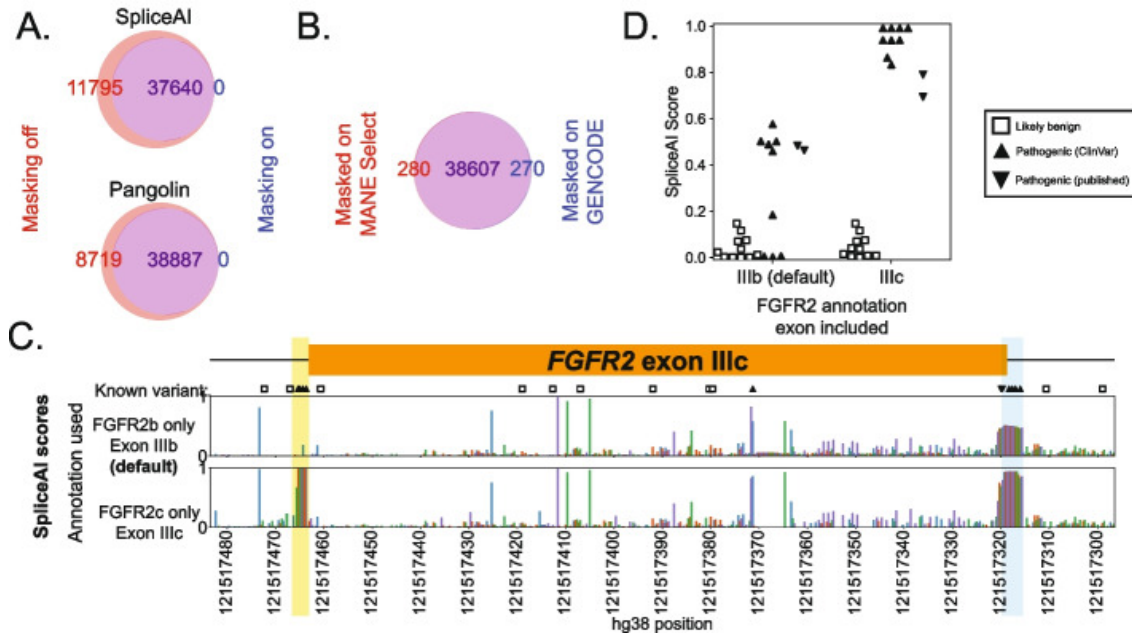


Figure 2: Masking and annotation change specialist calls. Among 500,000 background variants, turning masking on removed 11,795 SpliceAI calls and 8,719 Pangolin calls (panel A; counts read from the figure). Switching Pangolin between MANE Select and GENCODE annotation left 280 and 270 calls unique to each (panel B). Panels C and D show an FGFR2 exon whose scores depend on the annotated isoform. These are background calls, not MFASS accuracy. Source: Smith and Kitzman, *Genome Biology* 2023, Fig. 6 [3], <https://pmc.ncbi.nlm.nih.gov/articles/PMC10734170/figure/Fig6/>, CC BY 4.0, unmodified.

3.2 Common scored population and exclusions

All four specialist configurations scored the same 8,297 of the 8,324 held-out variants (314 disrupting, 460 groups). Every metric uses this common population; the 27 unscored variants are excluded, not scored as negatives. One excluded variant is a disruptor. The exclusions addendum of the benchmark [12] gives two causes:

- **23 assembly-orientation exclusions.** The recorded alleles are the complement of the GRCh38 base, consistent with an inverted hg19-to-hg38 mapping whose strand and alleles were not converted. Both tools refused these variants. The addendum traces 23 held-out records (21 from one CYFIP1 exon) and 90 cohort records to this cause; 67 are in the training arm. Another AI coding agent (Codex) checked the explanation computationally and it was reported upstream [13]; the MFASS authors have not confirmed it.
- **4 annotation-span exclusions.** Four ARHGEF3 variants fall outside the GENCODE 44 canonical transcript span, a consequence of the frozen transcript choice, not faulty variants.

3.3 Configurations

Table 1 lists the ten configurations. Three evidence types are distinguished: *computed or fitted here* (scores produced by the companion’s controls step on the training arm); *saved-score replay* (published per-variant SpliceAI and Pangolin predictions from the annotation-matched study [14], re-evaluated without rerunning either tool); and specialist inference, which was *not run* for this work.

3.4 Controls and supervised baselines

prevalence assigns the training prevalence (0.0379) to every variant and serves as a no-information floor. **distance** scores each variant by the negative distance, in bases, to the nearest exon–

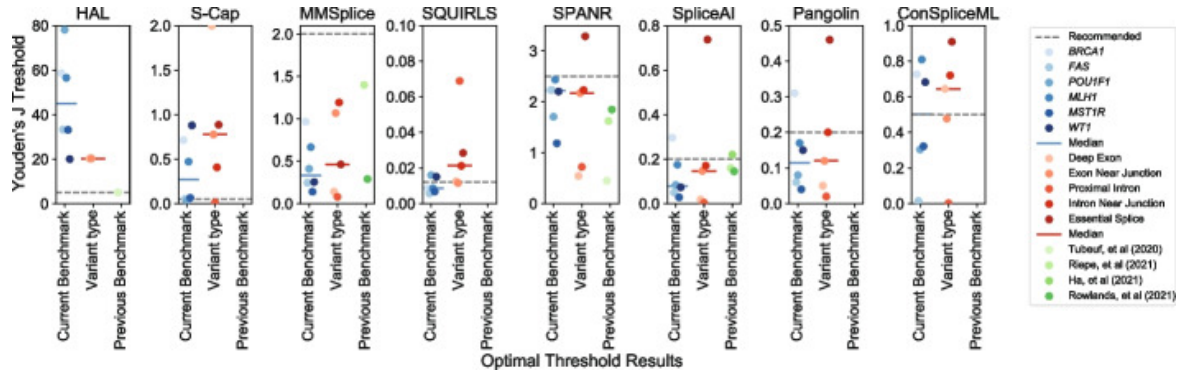


Figure 3: Optimal thresholds vary by dataset and variant type. Youden-optimal thresholds for eight predictors (points) by dataset, variant type and earlier report, against the benchmark’s recommended threshold (dashed line; 0.2 for SpliceAI and Pangolin). HAL and ConSpliceML optima span nearly the whole score range; SpliceAI and Pangolin optima vary less and mostly fall below 0.2. None of the datasets is MFASS. Source: Smith and Kitzman, *Genome Biology* 2023, Fig. 5 [3], <https://pmc.ncbi.nlm.nih.gov/articles/PMC10734170/figure/Fig5/>, CC BY 4.0, unmodified.

intron junction in the construct (0 is the base touching a junction); it is a fixed rule with no fitting. The two k -mer models reproduce the published MFASS-v2 baseline features (following `run_baseline.py` in `rewire-benchmarks`, MIT licence): position of the variant relative to both junctions and the construct, exon and intron lengths, region and allele indicators, 3-mer frequencies in a 21 bp window of the assay-oriented mutant sequence, and, for `kmer_cons`, phyloP and mean phastCons scores. Both use scikit-learn’s histogram gradient-boosted classifier (300 iterations, L2 regularisation 1.0, seed 20260914) with the published settings (learning rate 0.06, 31 leaves). The companion’s refit of `kmer_cons` reproduces the published baseline predictions byte for byte (SHA-256 2f3117c2...23eb).

A validation-selected variant of each k -mer model was also fitted. A grid over learning rate $\{0.03, 0.06, 0.1\}$ and leaves $\{15, 31\}$ was evaluated by average precision on a 25% group-held-out part of the training arm (4,376 variants, 282 groups, 134 disrupting); the test arm is never read. The grid preferred 15 leaves for both models and learning rate 0.1 for `kmer_cons` (Appendix Table 14), so validation selection disagrees with the published settings. The historical baseline remains the default reference for every contrast; the selected versions appear only as a labelled sensitivity analysis. Selection used average precision, whereas the decision endpoint is P@100.

3.5 Specialist saved-score replay

The matched study [14–16] built both tools’ annotation from one `Ensembl_canonical` transcript per gene in the GENCODE 44 GTF [17] (the companion’s `annotation` step finds 62,754 such transcripts), used a scoring distance of 50 and patched Pangolin to give each gene its own score arrays before masking [18]; P0 and P1 are therefore not stock Pangolin. Its settings correspond to SpliceAI `-D 50 -M 0` or `-M 1` and Pangolin `-d 50 -m False` or `-m True`. The companion downloads the study’s per-variant predictions at commit 093fd1a and re-evaluates them; the replay reproduces the study’s AP and AUROC for all four conditions to 10^{-12} and its registered-order P@100 exactly. Matching the annotation does not isolate architecture: padding, masking rules, training data, ensemble size and score definition still differ.

3.6 Metrics

Each construct costs about the same whatever its result, so the primary metric is precision at budget k : the fraction of the first k ranked variants that disrupt splicing. The cost modelled is a construct that shows no defect; missed disruptors are not costed, although recall is recorded. A random 100 would hold about 4 disruptors. SpliceAI and Pangolin report two-decimal scores, so

Table 1: Configurations compared. “Fitted” means trained on the 19,409 training variants (1,127 groups) and scored locally by the companion `controls` command. “Replay” means published per-variant predictions from the annotation-matched study [14] at 093fd1a, re-evaluated here. Overlap between MFASS and the specialists’ training data was not audited.

Name	What it is	Inputs	MFASS labels	Scores
prevalence	Training prevalence (0.038) for every variant	None	One number	Computed
distance	Negative distance to the nearest exon–intron junction in the construct	Construct layout	No	Computed (fixed rule)
kmer_assay	Gradient-boosted trees on 3-mer frequencies in a 21 bp window, construct position and alleles; published settings	Construct sequence and layout	Yes	Fitted
kmer_cons	kmer_assay plus phyloP and phastCons; the historical MFASS-v2 baseline, published settings	As above plus conservation	Yes	Fitted
kmer*_selected	As above, hyperparameters chosen on held-aside training groups	As above	Yes	Fitted
S0 / S1	SpliceAI 1.3.1, 5 models, mask 0 / 1, distance 50	GRCh38 position and alleles; GENCODE 44 canonical annotation	No fitting	Replay
P0 / P1	Pangolin 5cf94b8 with a per-gene masking patch, 12 models, mask False / True, distance 50	As for SpliceAI	No fitting	Replay

ties at the cut-off are common and are handled explicitly. Let c be the k -th highest score, A the set of variants scoring above c and T the set tied at c , with p_A, p_T disrupting variants and n_A, n_T variants in each; $r = k - n_A$ tied slots remain. Then

$$\mathbb{E}[\text{P@}k] = \frac{p_A + r p_T / n_T}{k}, \quad \min = \frac{p_A + \max(0, r - (n_T - p_T))}{k}, \quad \max = \frac{p_A + \min(r, p_T)}{k}, \quad (1)$$

where the expectation is over a uniformly random order of the tied block (Appendix B lists the companion function verbatim). Paired contrasts instead use the *registered tie order*: the matched study’s fixed, label-independent permutation of the variants. Average precision (AP; non-interpolated, scikit-learn [19]) and AUROC summarise the whole ranking, which matters when the budget is large or open.

3.7 Paired bootstrap and interval inventory

Intervals are 95% percentile intervals from 2,000 whole-group bootstrap resamples (seed 20260914) of the 460 exon/gene groups, computed on the candidate-minus-baseline difference. Because a resampled list is longer or shorter than the original, each draw keeps the review *fraction* $k/8,297$ of its list, rather than exactly k variants; the realised depth was 21–31 for a nominal 25, 83–126 for 100 and 250–378 for 300 (Appendix Table 16). The observed difference uses exactly k . No draw was single-class.

In this manuscript “no difference established” means that a paired 95% interval for the

difference includes zero. That is not equivalence: such an interval can also include differences large enough to matter. The benchmark protocol’s improvement rule (a P@100 gain of at least 0.05 with a paired lower bound strictly above zero [20]) applies only to a future, prospectively registered cohort and is not applied here either way. Interval bounds within about 0.005 of zero are described as “at or near zero”; the Monte Carlo error of the bootstrap endpoints was not quantified.

The study reports 41 substantive intervals, none adjusted for multiplicity: 9 from the matched study, all between specialists (S1–S0, P1–P0 and P0–S0 on P@100, AP and AUROC); 20 distinct intervals from the companion run against `kmer_cons` (four specialists $\times$ P@25, P@100, P@300, AP and AUROC; AP and AUROC repeat identically at each budget); and 12 against `kmer_cons_selected`. The 12 tutorial-subset intervals in Appendix Table 17 are a pipeline check and are neither counted nor interpreted.

3.8 Pre-specified and post hoc analyses

Only $k = 100$ was fixed in advance. Budgets 25 and 300 were chosen after the budget curve (Figure 4) had been seen, and the tables mark them post hoc. The validation-selected baseline contrasts come from a separate run executed after the analyses above. The junction-distance breakdown and the shortlist-overlap counts are descriptive outputs that `evaluate` prints for every configuration; they were not pre-specified as endpoints and carry no intervals.

3.9 Scoring checks

The companion’s `check-variant` command runs the checks that decide whether a variant can be scored on GRCh38: (1) the assay pair; (2) the GRCh38 base against the recorded genomic reference allele; (3) the 170 bp assay reference window against GRCh38 on the recorded strand; and (4) the GENCODE 44 canonical transcript and distance to the nearest canonical exon end. It prints the public GRCh38 base but never the cohort’s alternate allele, because the MFASS repository declares no licence. Three worked examples, recorded in the original run (Appendix C), illustrate the outcomes:

- A DDX1 intronic variant (ENSE00000712808_003) passes checks 1–3 with 0 mismatching bases, lies 1 bp from a canonical exon end and tops the unmasked Pangolin shortlist (S0 and S1 1.0000; P0 and P1 0.9100; outcome disrupting). Its outcome was known beforehand, so this is a worked example, not validation.
- A CYFIP1 variant (ENSE00001321140_001) fails check 2: the recorded alleles are the complement of the GRCh38 base, the window has 135 mismatching bases on the recorded strand and 0 on the opposite strand. The assay is intact; the mapping inverted the region. It is excluded, not repaired.
- An ARHGEF3 variant (ENSE00002361772_001) passes checks 1–3 but lies about 80 kb beyond the canonical transcript (ENST00000296315.8), although ten alternative transcripts cover it; it is unscored under the canonical-transcript protocol.

These checks matter because the 23 orientation mismatches surfaced only because both tools refused them; a pipeline that skipped the reference-allele check, or flipped mismatched alleles automatically, would have scored them without warning. A mismatched base should not be complemented unless the mapping orientation and sequence context have been validated.

3.10 Execution and provenance

Every companion command reported in the original article, other than one illustrative example marked as not executed, was executed on 29 and 30 September 2026 by an AI coding agent

Table 2: Budget 100 on the common 8,297 variants (314 disrupting, 460 groups). P@100 is the expected value over random orders of the tied block, with the worst and best cases in brackets; “Reg.” is P@100 under the registered tie order; “Scored” counts variants scored of the 8,324 held out; “Tied” is the number of variants tied at the cut-off score. Recall is the expected recall at 100. Source: archived out/metrics.json, methods.

Configuration	Scores	Scored	P@100 [tie range]	Reg.	Tied	Recall	AP	AUROC
prevalence	Computed	8,324	0.038 [0.00, 1.00]	0.04	8,297	0.012	0.038	0.500
distance	Computed (fixed rule)	8,324	0.135 [0.00, 0.23]	0.11	171	0.043	0.066	0.658
kmer_assay	Fitted	8,324	0.570 [0.57, 0.57]	0.57	1	0.182	0.274	0.787
kmer_cons	Fitted	8,324	0.610 [0.61, 0.61]	0.61	1	0.194	0.287	0.778
kmer_assay_selected	Fitted (val. selected)	8,324	0.580 [0.58, 0.58]	0.58	1	0.185	0.273	0.780
kmer_cons_selected	Fitted (val. selected)	8,324	0.630 [0.63, 0.63]	0.63	1	0.201	0.298	0.778
S0	Replay	8,297	0.630 [0.63, 0.63]	0.63	1	0.201	0.295	0.804
S1	Replay	8,297	0.650 [0.65, 0.65]	0.65	2	0.207	0.313	0.815
P0	Replay	8,297	0.650 [0.65, 0.65]	0.65	1	0.207	0.389	0.876
P1	Replay	8,297	0.657 [0.65, 0.66]	0.66	3	0.209	0.411	0.873

(Claude) on an Apple M4 CPU (16 GB, macOS 26.6.2, uv 0.8.2, Python 3.11.16, NumPy 1.26.4, scikit-learn 1.9.1). The companion fetches 14 pinned files with SHA-256 checks: the MFASS tables (about 80 MB) from the authors’ repository at [9a8e4f2](#) and the split and saved predictions (about 2 MB) from rewire-benchmarks at [093fd1a](#). The final revision was rerun from the archive in an empty directory with a fresh uv cache and fresh downloads: every command exited 0, the decision tables, shortlists, exclusion lists and control predictions were byte-identical, `metrics.json` differed only in timing fields, and the test suite reported 26 passed. Those statements are part of the historical record; they were not re-executed for this manuscript. The tables below are formatted from the archived run outputs (`downloads/splice-shortlist-outputs.zip`, SHA-256 prefix `6d1f0de2f359`); each archive member’s digest is checked before use. Every generated value that also appears in the original article (evidence-map items T2–T6, C-POP, C-OVERLAP, F4 and M-STUDY) is cross-checked against it; the remaining values are formatted directly from the digest-checked archive members (Appendix F).

4 Results

4.1 At a budget of 100, no specialist configuration was separated from the baseline

Table 2 summarises all configurations at $k = 100$. The historical baseline `kmer_cons` reached P@100 0.610, the four specialist configurations 0.630 to 0.657, and the validation-selected baseline 0.630, the same point estimate as unmasked SpliceAI. The `distance` rule’s 0.135 is an expectation over a large tie: 171 variants share its top score (23 disrupting), so its first 100 can hold 0 to 23 disruptors.

Table 3 gives the paired differences against `kmer_cons`. At 100, no P@100 difference interval excludes zero; together they run from -0.069 to $+0.133$, which includes both a meaningful loss and a meaningful gain. Under the least favourable tie order, P1’s $+0.050$ would be $+0.040$.

4.2 Whole-list metrics favour Pangolin (exploratory, unadjusted)

The whole-list metrics differ more than P@100. Both Pangolin configurations have AP and AUROC intervals above zero against the historical baseline (unmasked: AP $+0.102$ [$+0.061$, $+0.142$], AUROC $+0.099$ [$+0.073$, $+0.125$]). Masked SpliceAI’s AUROC interval against the historical baseline, $+0.037$ [$+0.005$, $+0.070$], excludes zero only marginally (lower bound at or

Table 3: Paired differences against the historical baseline `kmer_cons`. Observed difference (candidate – `kmer_cons`) with the 95% whole-group bootstrap interval in brackets beneath it (2,000 draws, seed 20260914, unadjusted, exploratory). P@k differences use the registered tie order; at 300, tie order moves P0 by -0.017 to $+0.007$ and P1 by -0.007 to $+0.010$. AP and AUROC do not depend on the budget. Budgets 25 and 300 are post hoc. Each draw keeps the review fraction $k/8,297$ (realised depths in Appendix Table 16). Source: archived `out-b25/`, `out/` and `out-b300/metrics.json`, `paired_contrasts`.

Candidate	P@25 (post hoc)	P@100	P@300 (post hoc)	AP	AUROC
S0	-0.040 [$-0.250, +0.208$]	$+0.020$ [$-0.069, +0.100$]	$+0.017$ [$-0.030, +0.070$]	$+0.008$ [$-0.039, +0.058$]	$+0.026$ [$-0.008, +0.059$]
S1	-0.040 [$-0.250, +0.208$]	$+0.040$ [$-0.046, +0.120$]	$+0.030$ [$-0.016, +0.088$]	$+0.026$ [$-0.021, +0.077$]	$+0.037$ [$+0.005, +0.070$]
P0	$+0.080$ [$-0.120, +0.304$]	$+0.040$ [$-0.038, +0.120$]	$+0.127$ [$+0.080, +0.172$]	$+0.102$ [$+0.061, +0.142$]	$+0.099$ [$+0.073, +0.125$]
P1	$+0.080$ [$-0.120, +0.304$]	$+0.050$ [$-0.026, +0.133$]	$+0.150$ [$+0.099, +0.199$]	$+0.124$ [$+0.083, +0.165$]	$+0.095$ [$+0.067, +0.125$]

Table 4: Paired differences against the validation-selected baseline `kmer_cons_selected` at $k = 100$. Same population, bootstrap and seed as Table 3 (realised depths 83–126); run after the main analyses. Source: archived `out-selected-b100/metrics.json`, `paired_contrasts`.

Candidate	P@100	AP	AUROC
S0	$+0.000$ [$-0.086, +0.101$]	-0.002 [$-0.047, +0.048$]	$+0.026$ [$-0.009, +0.060$]
S1	$+0.020$ [$-0.063, +0.111$]	$+0.015$ [$-0.031, +0.065$]	$+0.037$ [$+0.003, +0.070$]
P0	$+0.020$ [$-0.046, +0.106$]	$+0.091$ [$+0.053, +0.131$]	$+0.098$ [$+0.073, +0.124$]
P1	$+0.030$ [$-0.033, +0.117$]	$+0.113$ [$+0.074, +0.154$]	$+0.095$ [$+0.065, +0.126$]

near zero; Monte Carlo error not quantified) among 41 unadjusted intervals, so it is not read as an established difference.

4.3 Sensitivity: the validation-selected baseline

The historical baseline is not the only defensible one. Table 4 repeats the contrasts at $k = 100$ against `kmer_cons_selected` (learning rate 0.1, 15 leaves), from a separate run after the analyses above with the same population, bootstrap and seed. The P@100 point differences shrink to between $+0.000$ and $+0.030$ and every P@100 interval still includes zero; across both baselines about 9 fewer to 13 more disruptions per 100 remain possible. The Pangolin AP and AUROC intervals still exclude zero; masked SpliceAI’s AUROC interval again excludes zero only marginally (lower bound $+0.003$, at or near zero).

4.4 Specialist-versus-specialist contrasts

The matched study reported nine intervals, all between specialists (Table 5). These values are cited from that study, not recomputed by the companion. None of the three specialist contrasts established a top-100 difference. The two masking contrasts have P@100 lower bounds of exactly 0.000 because P@100 moves in steps of about 0.01 and many draws have a difference of exactly zero (22.5% for SpliceAI masking, 44.4% for Pangolin masking). The P1–P0 P@100 difference also depends on tie order: any label-independent order gives P1 0.65 or 0.66. Masking raised AP within each tool without an established change in AUROC or P@100. Under matched annotation, unmasked Pangolin exceeded unmasked SpliceAI in AP and AUROC. No other specialist pair was measured.

Table 5: Specialist-pair contrasts reported by the annotation-matched study (8,297 common variants, 314 disrupting, 460 groups; whole-group bootstrap, 2,000 draws, seed 20260914, 95% percentile intervals; each draw keeps the review fraction, realised depths 83–126). The original article quoted five of these nine intervals; the remaining four (both masking P@100 upper bounds and both masking AUROC intervals) and the P1–P0 P@100 point value are transcribed from the study’s published README and contrast files at 093fd1a [14] without recomputation (`evidence/paper-migration/matched-study-contrasts.json`).

Contrast	P@100	AP	AUROC
S1 – S0 (SpliceAI masking)	+0.020 [+0.000, +0.062]	+0.017 [+0.006, +0.028]	+0.011 [−0.007, +0.027]
P1 – P0 (Pangolin masking)	+0.010 [+0.000, +0.040]	+0.022 [+0.011, +0.034]	−0.004 [−0.019, +0.013]
P0 – S0 (unmasked tools, matched annotation)	+0.020 [−0.033, +0.083]	+0.093 [+0.064, +0.124]	+0.073 [+0.048, +0.096]

Table 6: Expected precision at each tested budget with the range over tie orders on the 8,297 common variants. Point estimates only; paired intervals are in Table 3. Budgets 25 and 300 are post hoc. Source: archived `methods` blocks of `out-b25/`, `out/` and `out-b300/metrics.json` (identical to their `decision-table.md`).

Configuration	P@25 [tie range]	P@100 [tie range]	P@300 [tie range]
<code>distance</code>	0.135 [0.00, 0.92]	0.135 [0.00, 0.23]	0.111 [0.08, 0.12]
<code>kmer_assay</code>	0.800 [0.80, 0.80]	0.570 [0.57, 0.57]	0.310 [0.31, 0.31]
<code>kmer_cons</code>	0.720 [0.72, 0.72]	0.610 [0.61, 0.61]	0.333 [0.33, 0.33]
<code>kmer_assay_selected</code>	0.720 [0.72, 0.72]	0.580 [0.58, 0.58]	0.310 [0.31, 0.31]
<code>kmer_cons_selected</code>	0.840 [0.84, 0.84]	0.630 [0.63, 0.63]	0.347 [0.35, 0.35]
<code>S0</code>	0.715 [0.68, 0.84]	0.630 [0.63, 0.63]	0.350 [0.35, 0.35]
<code>S1</code>	0.715 [0.68, 0.84]	0.650 [0.65, 0.65]	0.366 [0.36, 0.37]
<code>P0</code>	0.817 [0.80, 0.84]	0.650 [0.65, 0.65]	0.457 [0.44, 0.47]
<code>P1</code>	0.817 [0.80, 0.84]	0.657 [0.65, 0.66]	0.484 [0.48, 0.49]

4.5 The answer depends on the budget

Figure 4 shows expected precision at the ten recorded budgets from 10 to 800 (values in Appendix Table 11). As fractions of this list, 25, 100 and 300 are 0.3%, 1.2% and 3.6% of 8,297 variants, against a prevalence of 3.8%. The curves overlap from about 25 to 150; from about 200 the two Pangolin curves are highest. SpliceAI is low at $k = 10$ (0.548) because only three of its seven variants at score 1.00 are disrupting. Table 6 gives the tie ranges at the three tested budgets and Figure 5 the paired differences.

Budget 25 (post hoc). The four measured difference intervals each span more than 0.4 and all include zero (Table 3). No other pair was tested at 25, including the baseline against its validation-selected version. Point estimates are fragile: `kmer_cons` finds 18, `kmer_assay` 20, `kmer_cons_selected` 21, Pangolin 20 or 21 and SpliceAI 17 to 21. Hyperparameters alone move the baseline from 0.720 to 0.840, more than any specialist point difference.

Budget 300 (post hoc). Expected yields are about 100 for `kmer_cons`, 105 and 110 for SpliceAI, and 137 (133 to 140 by tie order) and 145 (143 to 148) for P0 and P1. The two Pangolin difference intervals exclude zero, +0.127 [+0.080, +0.172] and +0.150 [+0.099, +0.199]; the SpliceAI intervals do not. These are the only budget-based contrasts whose intervals exclude zero, and both rest on a post hoc budget. The validation-selected baseline was not tested at 300.

4.6 Ties and shortlist composition

Ties are a separate problem from scoring checks. At $k = 25$, both SpliceAI configurations have 7 variants at 1.00 (3 disrupting) and 23 tied at 0.99 (19 disrupting) for the remaining 18 slots, so

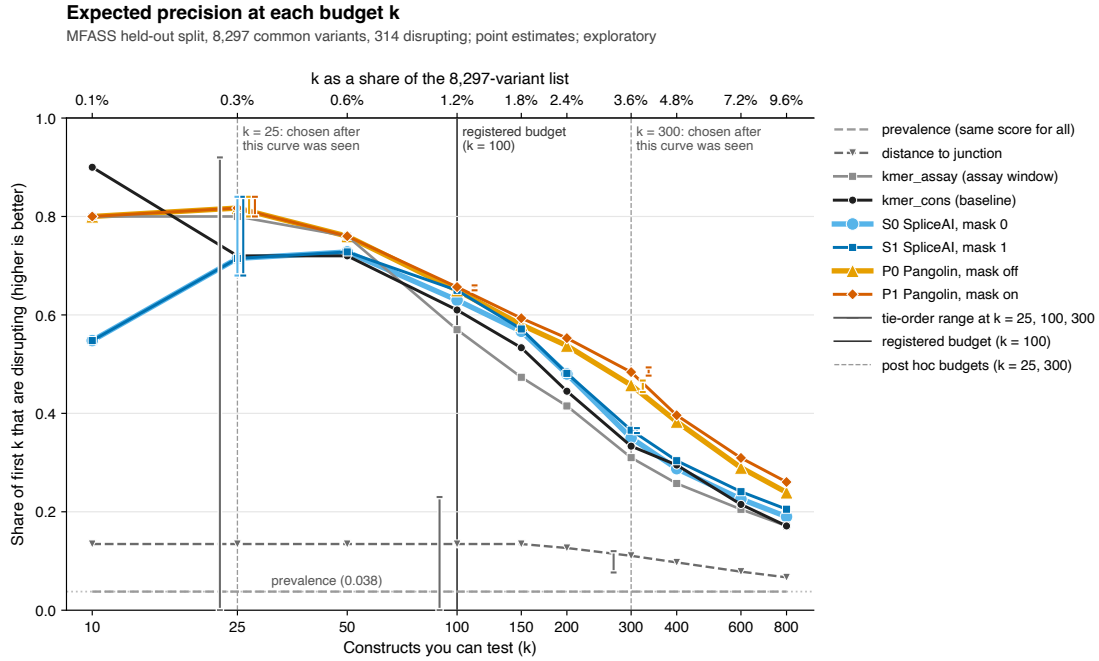


Figure 4: Expected precision at each budget k . Lines show expected $P@k$ on the 8,297 common variants; bars at 25, 100 and 300 are tie-order ranges. The tall grey bars belong to the **distance** rule (0.00 to 0.92 at 25); **prevalence** has no bar because every variant ties. Budgets 25 and 300 were chosen after this curve had been seen. Source: original figure `article/assets/04-budget-curve.svg` (converted to PDF), drawn from `out/metrics.json methods.<name>.budget_curve` and the `precision_at_budget_range` fields of the three budget runs.

the first 25 hold 17 to 21 disrupting variants ($P@25$ from 0.68 to 0.84). The registered tie order happened to give 17, which is why SpliceAI’s $P@25$ contrast in Table 3 is -0.040 . The exported budget-25 shortlist therefore has 30 rows, 23 of them flagged as tied with the cut-off; its rank within a tied block is arbitrary. At 100, masked Pangolin has 3 variants tied at 0.54 for 2 slots, one disrupting, so its $P@100$ is 0.65 or 0.66. A tie rule should be fixed before any outcome is seen, such as a seeded random permutation, or the whole tied block should be tested.

Under the registered tie order, the first 100 of **kmer_cons** and P0 share only 61 variants, and S0 and P0 share 88 (all pairs in Appendix Table 13). Two disrupting variants illustrate the disagreement (reported in the original article from the companion run; their identifiers are not in the archived outputs, so these values are imported and not independently re-checked). An IPO9 disrupting variant 3 bp from an exon end is ranked 1,092nd by **kmer_cons** and 81st to 87th by the four specialists. A COL1A2 disrupting variant, also 3 bp from an exon end, is ranked 95th by **kmer_cons**; unmasked SpliceAI scores it 0.00, one of 4,448 variants tied at ranks 3,850 to 8,297, and unmasked Pangolin 0.05, one of 182 tied across ranks 795 to 976.

4.7 Where the disruptors are: junction-distance bands

Table 7 breaks the captured disruptors down by distance to the nearest exon–intron junction in the construct. At 100, most captured disruptors lie within 10 bases of a junction (51 of 61 for **kmer_cons**, 57 of 65 for P0), and **kmer_cons** captures slightly more at 11–30 bases (10 against 8). No top 100 reaches the 30 disruptors beyond 30 bases, although 173 of the 314 lie more than 10 bases out. At 300, 36 of P0’s 38 extra disruptors over **kmer_cons** (43 of 45 for P1) are 3 or more bases from a junction. Pangolin reaches 5 or 6 distant disruptors against none for the k -mer models: small counts at a post hoc budget. No band intervals were computed. Appendix Table 12 gives every configuration at all three budgets.

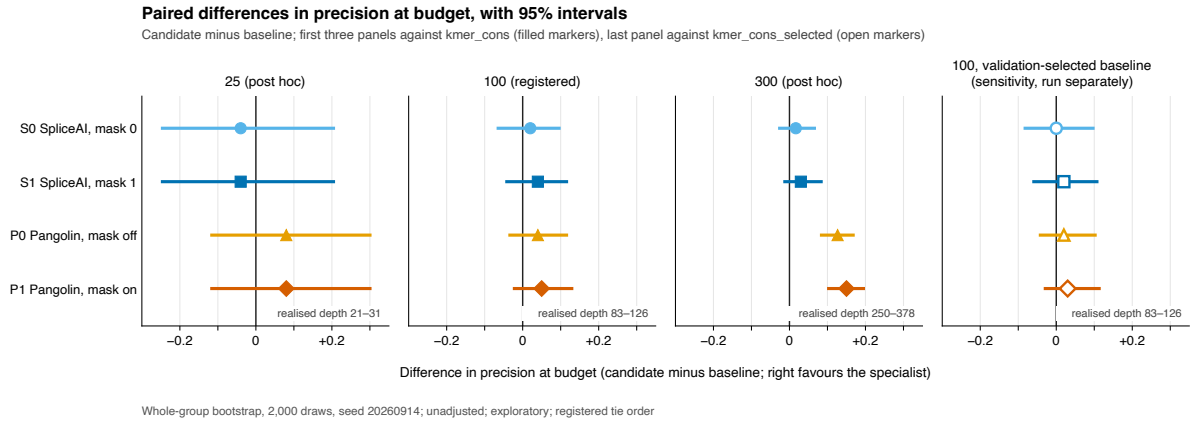


Figure 5: Paired P@k differences with 95% whole-group bootstrap intervals. Panels show S0, S1, P0 and P1 minus the historical baseline at budgets 25, 100 and 300, and a separately run panel at 100 against the validation-selected baseline. At 25 and 100 every interval includes zero (at 100, against either baseline). At 300 (post hoc) both Pangolin intervals exclude zero. Source: original figure [article/assets/05-paired-differences-forest.svg](#) (converted to PDF), drawn from the `paired_contrasts` of `out-b25/`, `out/`, `out-b300/` and `out-selected-b100/metrics.json`.

Table 7: Disrupting variants captured in each top 100 and top 300 (registered tie order), by distance to the nearest exon–intron junction in the 170 bp construct (0 is the base touching a junction), with within-band AUROC at the right. Descriptive, not pre-specified; the top-300 columns inherit the post hoc status of that budget; no band intervals were computed. Source: archived `out/metrics.json` and `out-b300/metrics.json`, `junction_distance_bands`.

Band (bases)	Variants	Disrupting	Top 100			Top 300 (post hoc)			AUROC in band
			<code>kmer_cons</code>	P0	P1	<code>kmer_cons</code>	P0	P1	
0–2	560	46	24	25	25	29	31	31	0.83 / 0.90
3–10	1,678	95	27	32	33	46	58	60	0.77 / 0.88
11–30	4,223	143	10	8	8	25	44	48	0.75 / 0.86
>30	1,836	30	0	0	0	0	5	6	0.75 / 0.84

4.8 Resources and coverage

The controls are cheap: loading, the 12-fit validation grid, final fits and scoring of 8,324 rows took 30.81 s end to end with 278.1 MB peak resident memory on the Apple M4 (the separate `build` step peaks at about 680 MB), and need only the MFASS tables (about 80 MB), which include conservation scores. The specialist times in Table 8 are the matched study’s recorded times for one sequential run on the same CPU type with 5 (TensorFlow) or 6 (Torch) threads, excluding cohort preparation; peak memory was not recorded. The study recorded about 14 h of wall time for the four conditions. Running a specialist yourself requires about 1.13–1.14 GB of downloads (Appendix Table 10). The specialists scored 8,297 of 8,324 held-out variants; archived runs with other annotations scored 8,194 and 8,301 and are not pooled here [14].

4.9 Reproduction status

Not yet reproduced under this repository’s harness. The values above are imported from the 2026-09-29/30 companion run. The repository’s reproduction protocol (`protocol.md`, draft v2, unapproved) incorporates corrections from an automated AI methods review of draft v1; draft v2 itself has not been reviewed, approved or executed, and no reproduction outcome class is reported. When a run is approved and ends, its results are to be shown next to these imported values, including any discrepancies, rather than replacing them.

Table 8: Recorded compute. Controls were measured in the companion run; specialist scoring times were recorded by the matched study (`score_test` covers 8,324 held-out rows). Peak memory for the specialists was not recorded. Source: archived `out/metrics.json, resources; runs/controls/receipt.json`.

Step	Evidence	Time	Peak memory
Controls (all six, incl. 12-fit validation grid)	Measured in the companion run	30.81 s end to end	278.1 MB
S0 scoring of the held-out arm	Recorded by matched study	6,921 s (0.83 s/variant)	not recorded
S1 scoring of the held-out arm	Recorded by matched study	6,102 s (0.73 s/variant)	not recorded
P0 scoring of the held-out arm	Recorded by matched study	19,153 s (2.30 s/variant)	not recorded
P1 scoring of the held-out arm	Recorded by matched study	18,655 s (2.24 s/variant)	not recorded

5 Discussion

5.1 What the evidence supports

At the one pre-specified budget, roughly 1.2% of this list, the specialists’ point estimates were 2 to 5 disruptors per 100 higher than the historical supervised baseline and 0 to 3 higher than its validation-selected version, but no difference was established and the intervals are wide enough to include a meaningful loss. Both supervised baselines need same-design labels, and `kmer_cons` also needs conservation tracks (it scored every variant in about 30 s with no genome download, but a new list needs both the tracks and thousands of labels); `kmer_assay` (P@100 0.570) needs no conservation. The specialists need GRCh38 coordinates, allele and strand checks and a canonical annotation. Choosing between them at this budget is therefore reasonably made on cost, coverage and the availability of labels rather than on the measured precision.

Whole-list metrics tell a more consistent story for Pangolin: higher AP and AUROC than both baselines and than unmasked SpliceAI under matched annotation, and, at the post hoc budget of 300, precision differences against the historical baseline whose intervals exclude zero. Because 300 was chosen after the curve was seen and none of the 41 intervals is adjusted, this is a lead to test prospectively, not a finding.

5.2 Decision guidance

Table 9 summarises where to start, by label availability and budget fraction; Appendix Figure 6 draws the same routes as a flowchart. Budgets are fractions of this one list: 8,297 MFASS variants with 3.8% prevalence and MFASS’s short-exon case mix. A list with a different size, case mix or assay needs its own labelled validation before these results guide the choice, so each row says where to start a pilot, not where to set a threshold.

5.3 What would change these conclusions

A prospectively registered comparison on an MFASS-like library with unseen outcomes, and with budget, tie rule, comparator and candidate fixed in advance, would replace the exploratory status of every row. If a specialist met the protocol’s improvement rule [20] there against a baseline trained on that library, it would become the recommendation where labels exist. Without labels, the deciding contrast is Pangolin against SpliceAI, tested the same way. If Pangolin’s lead at about 3.6% of the list did not reappear, the advice to pilot Pangolin first at that fraction would be withdrawn. An author-confirmed liftover correction would return the 23 exclusions and change the denominator. Patient RNA outcomes would answer a different question.

Table 9: Where to start, by budget fraction and label availability (MFASS-like minigene endpoint). “Same-design labels” are assay labels from the same construct design. Budgets are fractions of this list of 8,297 variants, not thresholds for other lists.

Situation	Where to start	What the evidence shows	Main caveat
Same-design labels; about 1.2% of the list (tested at 100)	Either baseline or any specialist; choose on cost and coverage (Table 2)	No P@100 interval excludes zero, against either baseline (Tables 3, 4)	Not equivalence: about 9 fewer to 13 more disruptions per 100 remain possible
No assay labels; about 1.2% (tested at 100)	A specialist, after GRCh38 allele, strand and transcript checks; choose on runtime, downloads and licence	distance finds about 13 per 100, the specialists 63 to 66 (Table 2); no specialist-pair top-100 difference established (Table 5)	distance versus specialists was not tested; P0–S0 allows about 3 fewer to 8 more per 100
Same-design labels; about 3.6% (tested at 300, post hoc)	Pilot Pangolin (P0 or P1)	Both Pangolin P@300 intervals exclude zero against the historical baseline (Tables 3, 7)	Post hoc budget; validation-selected baseline not tested at 300
Same-design labels; about 0.3% (tested at 25, post hoc)	Fix the tie rule before testing	No measured interval excludes zero (Table 3); ties move SpliceAI by up to 4 of 25 and hyperparameters move the baseline by 3 (Table 6)	Post hoc budget; point estimates are fragile
No labels at any fraction other than about 1.2%; labels at any fraction other than about 0.3, 1.2 or 3.6%	Pilot before committing; without labels, start with a specialist	Point estimates only (Figure 4); without labels, P0 had higher AP and AUROC than S0 (exploratory)	No P@k contrast was tested at these fractions
Every variant must be scored	Expect losses from coordinate and annotation mapping	Specialists scored 8,297 of 8,324	Losses on another list depend on its mapping to GRCh38 and canonical transcripts
Patient RNA, pathogenicity or another reporter	None of these routes transfers	—	Pilot a labelled subset in the target assay

6 Limitations

- **Exploratory status and budgets.** The held-out outcomes had been inspected before, so every interval is exploratory and unadjusted for the 41 intervals reported. Only $k = 100$ was set in advance; 25 and 300 are post hoc, and the conclusion changes between 100 and 300. Every P@k figure applies to this fixed population and case mix.
- **Monte Carlo error.** The bootstrap endpoints’ Monte Carlo error was not quantified. Statements resting on bounds at or near zero (masked SpliceAI AUROC lower bounds +0.005 and +0.003; masking P@100 lower bounds 0.000) are fragile.
- **Coverage.** The specialists need a GRCh38 position, matching alleles and an annotated gene; they skipped 27 of 8,324 variants, including one disruptor.
- **Assembly.** The 23 orientation mismatches surfaced only because both tools refused them. The explanation is not author-confirmed [13]. The training arm holds 67 records with the same defect [12], whose effect on the k -mer baselines was not assessed.
- **Ties.** At 25, SpliceAI’s tie range (0.68 to 0.84) is wider than most differences between configurations.
- **Reporter to patient.** MFASS uses short in-frame exons with truncated introns, one cell line for the main screen, and counts alternative 5’ or 3’ splice-site usage as a false negative. In full-length genes the authors confirmed 13 of 19 tested variants, and five detected changes involved alternative splice sites [4]. Predicting a splicing assay “cannot directly imply clinical significance” [9]. The results say nothing about splicing in patient RNA, pathogenicity or diagnostic yield.

- **Labels, fairness and replay.** `kmer_cons` needs same-design labels and moves with its hyperparameters. The specialists are zero-shot on this assay but pretrained on other data whose overlap with MFASS was not audited, which could bias in their favour by an unknown amount. A specialist rerun with other versions or annotations may give different scores; specialist inference was not re-executed for this work, and its peak memory was never measured.
- **Imported case values.** The IPO9 and COL1A2 case ranks, the build counts, the canonical-transcript count and the `check-variant` outputs exist only as article text from the original run, not as archived machine outputs.
- **Platform.** Only macOS arm64 runs at default thread counts are recorded; determinism on other platforms or thread caps is untested.
- **Review.** The study, the exclusions and this analysis, including the companion’s UCSC-based assembly check, were checked only by automated agents (Claude, Codex), not by independent human scientific review.

7 Data, code and licence availability

The study repository (`rewire-bio/splicing-model-selection`) contains this manuscript and its build (`make paper-imported`), the original article (`article/original.md` and the byte-identical `article/published-original.md`, also embedded in this PDF as Supplement S1), the runnable companion (`companion/`, also as `downloads/splice-shortlist.zip`), the archived run outputs (`downloads/splice-shortlist-outputs.zip`) and the provenance records under `evidence/`.

The companion contains no MFASS data, model weights or third-party source; `fetch` downloads the MFASS tables from the authors’ repository and the saved predictions from rewire-benchmarks (MIT) at pinned commits. The archived outputs contain per-variant scores and predictions, MFASS outcome labels for the held-out variants, gene names and GRCh38 positions; they contain no MFASS source tables, assay sequences, alleles or genome files. MFASS declares no licence [10], so its tables are not redistributed and the redistribution status of the MFASS-derived labels is an open owner decision; no permission is inferred. The companion’s own licence is to be set by the author at publication. Pangolin is GPL-3.0, with no separate terms for its weights [7]. The current SpliceAI licence puts code under PolyForm Strict 1.0.0 and models under CC BY-NC 4.0, a change made in July 2025, while the PyPI metadata for 1.3.1 says GPLv3 [6, 21]; these sources do not settle which terms govern a given copy. The three reproduced literature figures are CC BY 4.0 and are attributed in their captions.

8 AI assistance, review and disclosure

The matched study, exclusions analysis and companion code come from this project’s own benchmark, rewire-benchmarks [11] (093fd1a). An AI coding agent (Claude) ran the companion commands, and the original article was drafted with the same model. The matched study had an automated Claude review [22]. Another AI agent (Codex) checked the assembly-orientation explanation. The repository’s reproduction protocol (`protocol.md`) and its methods review (`migration-review/review.md`) were also written by AI agents. This manuscript was drafted by an AI model (Claude Opus, `claude-opus-5-5`) from the archived evidence and was reviewed by an independent AI reviewer instance of the same model; the review, with 19 findings and their disposition, is recorded in `evidence/paper-migration/review.md`. None of this work has had independent human scientific review.

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A Reproducing the historical run

The commands below are adapted from the original article (`unzip` and `cd` joined; trailing comments moved to prose; one long command continued with a backslash). They were executed on 29–30 September 2026 and were *not* re-executed for this manuscript. `uv sync --frozen` installs the locked versions (NumPy 1.26.4, scikit-learn 1.9.1, SciPy 1.17.1); `fetch` downloads 14 pinned files and stops on any SHA-256 mismatch. They need `uv` 0.8 or later [23] and access to `raw.githubusercontent.com`, `ftp.ebi.ac.uk` and `api.genome.ucsc.edu`; data downloads are about 82 MB for `fetch` and 50 MB for `annotation`.

```
unzip splice-shortlist.zip && cd splice-shortlist
uv sync --frozen
uv run --frozen python splice_shortlist.py fetch
uv run --frozen python splice_shortlist.py build
uv run --frozen python splice_shortlist.py controls
uv run --frozen python splice_shortlist.py annotation
uv run --frozen python splice_shortlist.py check-variant ENSE00000712808_003
uv run --frozen python splice_shortlist.py check-variant ENSE00001321140_001
uv run --frozen python splice_shortlist.py check-variant ENSE00002361772_001
uv run --frozen python splice_shortlist.py evaluate --tutorial --budget 20 \
    --draws 200 --method P0 --out out-tutorial
uv run --frozen python splice_shortlist.py evaluate --budget 100 --method P0 --out out
uv run --frozen python splice_shortlist.py evaluate --budget 300 --method kmer_cons \
    --out out-b300
uv run --frozen python splice_shortlist.py evaluate --budget 25 --method S1 --out out-b25
uv run --frozen python splice_shortlist.py evaluate --budget 100 \
    --baseline kmer_cons_selected --method kmer_cons_selected --out out-selected-b100
uv run --frozen pytest -q
```

`build` should print the following lines and exits if any count drifts:

```
source rows 32669; eligible 27733 (paper 27,733); disrupting 1050 (paper 1,050); exons in mutant
set 2198 (paper 2,198)
assay-pair checks passed for all 27733 variants; legacy sequence reverse-complemented in 7770
train 19409 variants (735 disrupting) in 1127 groups; test 8324 (315 disrupting) in 463 groups
```

`controls` fits the controls on training groups only and reports that validation selection disagrees with the published settings for both k -mer models. `annotation` downloads the GENCODE 44 GTF, checks its MD5 and should report 62,754 `Ensembl_canonical` transcripts. Each full `evaluate` run takes about a minute; at 100 it ends wrote `out/shortlist.csv` (100 rows for budget 100; 1 tied at the cutoff). `-budget` must be between 1 and the number of commonly scored variants (8,297; 977 with `-tutorial`) and `-draws` at least 100; `-method` only chooses which shortlist is written and `-baseline` sets the contrast reference. `excluded.csv` lists the 27 exclusions with reasons; `pytest` should report 26 passed. The companion evaluates MFASS only; for another list, apply the same allele, strand and transcript checks with your own tooling. A command such as `-budget 60 -method P1 -out out-b60` would export a Pangolin shortlist for a budget of 60; that command was not executed.

Specialist inference (not executed). To regenerate rather than replay the specialist predictions, follow the matched study’s results README at 093fd1a [14]: download the GENCODE 44 FASTA and GTF, apply `pangolin-5cf94b8-mask-per-gene-1.patch` to Pangolin at 5cf94b8, build the two locked environments and run the steps in `manifest-v1.json` [16]. On macOS arm64 the downloads are listed in Table 10. The study recorded about 14 h of wall time for the four conditions.

Table 10: Specialist downloads (macOS arm64) recorded in the archived `out/metrics.json`, `resources.specialist_downloads_bytes` (decimal megabytes). SpliceAI needs the FASTA, GTF, spliceai and TensorFlow wheels; Pangolin the FASTA, GTF, models directory and torch wheel.

Item	Size
GRCh38 primary assembly FASTA (GENCODE 44, gzip)	845 MB
GENCODE 44 primary-assembly GTF (gzip)	50 MB
spliceai 1.3.1 wheel (includes 5 models)	17 MB
tensorflow 2.21.0 wheel (cp311 macOS arm64)	223 MB
Pangolin repository models directory at 5cf94b8	184 MB
torch 2.2.2 wheel (cp311 macOS arm64)	60 MB

Rebuilding this manuscript. make paper-imported runs `scripts/build_paper.py` with the system Python and local TeX Live. It verifies the archive and member digests, formats the tables in `paper/generated/`, converts the original SVG figures with `rsvg-convert`, and runs `pdflatex`, `BibTeX` and `pdflatex` twice. It performs no experiment, data download or environment creation. The build receipt is `evidence/paper-migration/build-receipt.json`.

B The budget_precision function

The listing below is verbatim from lines 511–527 of the companion script `splice_shortlist.py`. Here `np` is NumPy, and `check_budget` raises a `ValueError` unless $1 \leq k \leq N$.

```
def budget_precision(labels, scores, k):
    """Precision in the top k, with ties at the cutoff handled explicitly.

    Returns the expectation under a uniformly random order of the tied block,
    the worst and best achievable values, and the tie size at the cutoff.
    """
    labels, scores = np.asarray(labels), np.asarray(scores, dtype=float)
    check_budget(k, len(scores))
    cut = np.sort(scores)[:k-1][k-1]
    above, tied = scores > cut, scores == cut
    pa, na = int(labels[above].sum()), int(above.sum())
    pt, nt = int(labels[tied].sum()), int(tied.sum())
    slots = k - na
    return {"k": k, "expected": (pa + slots * pt / nt) / k,
            "min": (pa + max(0, slots - (nt - pt))) / k, "max": (pa + min(slots, pt)) / k,
            "tied_at_cutoff": nt, "slots_in_tied_block": slots,
            "positives_in_tied_block": pt}
```

C Recorded check-variant outputs

Recorded in the original article from the companion run (ellipses as published).

```
$ uv run --frozen python splice_shortlist.py check-variant ENSE00000712808_003
ENSE00000712808_003 gene DDX1 (ENSG00000079785) split test group g0135
hg38 chr2:15629601 gene strand + construct region upstr_intron assay position 42/170
1 assay pair: reference and mutant differ only at position 42, alleles consistent with strand:
  PASS (checked in build)
  legacy `sequence` column orientation: reverse_complement
2 GRCh38 base at chr2:15629601 is G; matches the recorded genomic ref allele: PASS
3 170 bp assay reference window vs GRCh38 chr2:15629560-15629729 on recorded strand +: 0
  mismatching bases
4 transcript: ENST00000233084.8 DDX1 chr2:15591868-15631101 (+); in exon: False; distance to
  nearest canonical exon end 1 bp
5 saved scores (replay):
```

```

S0: 1.0000
S1: 1.0000
P0: 0.9100
P1: 0.9100
MFASS outcome (large-effect disruption, retrospective): 1

$ uv run --frozen python splice_shortlist.py check-variant ENSE00001321140_001
ENSE00001321140_001 gene CYFIP1 (ENSG00000068793) split test group g0890
hg38 chr15:22944977 gene strand + construct region upstr_intron assay position 14/170
...
2 GRCh38 base at chr15:22944977 is G; matches the recorded genomic ref allele: FAIL
recorded alleles are the complement of the GRCh38 base
3 170 bp assay reference window vs GRCh38 chr15:22944964-22945133 on recorded strand +: 135
mismatching bases
same window on the opposite strand (-), chr15:22944821-22944990: 0 mismatching bases
-> the assay is intact; the hg19->hg38 mapping inverted this region but the strand and
alleles were not converted. Excluded, not repaired, in the study.
4 transcript: ENST00000617928.5 CYFIP1 chr15:22867052-22980368 (-); in exon: False; distance
to nearest canonical exon end 38 bp
...

$ uv run --frozen python splice_shortlist.py check-variant ENSE00002361772_001
...
3 170 bp assay reference window vs GRCh38 chr3:56882244-56882413 on recorded strand -: 0
mismatching bases
4 transcript: no GENCODE 44 Ensembl_canonical transcript spans this position -> unscored under
the canonical-transcript protocol
...

```

D Supplementary tables

All values are formatted from the archived run outputs without recomputation.

Table 11: Expected precision across the full recorded budget curve (Figure 4). Source: archived out/metrics.json, methods.<name>.budget_curve.

Configuration	10	25	50	100	150	200	300	400	600	800
prevalence	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038	0.038
distance	0.135	0.135	0.135	0.135	0.135	0.126	0.111	0.097	0.078	0.067
kmer_assay	0.800	0.800	0.760	0.570	0.473	0.415	0.310	0.258	0.205	0.171
kmer_cons	0.900	0.720	0.720	0.610	0.533	0.445	0.333	0.295	0.215	0.171
kmer_assay_selected	0.800	0.720	0.760	0.580	0.460	0.415	0.310	0.247	0.193	0.165
kmer_cons_selected	0.900	0.840	0.760	0.630	0.547	0.470	0.347	0.275	0.205	0.174
S0	0.548	0.715	0.728	0.630	0.567	0.480	0.350	0.288	0.226	0.190
S1	0.548	0.715	0.728	0.650	0.571	0.481	0.366	0.304	0.241	0.205
P0	0.800	0.817	0.760	0.650	0.580	0.537	0.457	0.383	0.290	0.239
P1	0.800	0.817	0.760	0.657	0.593	0.553	0.484	0.396	0.309	0.261

Table 12: Junction-distance bands for every configuration at budgets 25, 100 and 300: disrupting variants captured in the top k (registered tie order), with within-band AUROC in parentheses (AUROC does not depend on k). Source: archived `junction_distance_bands` of the three budget runs.

k	Band	Disr.	distance	kmer_assay	kmer_cons	S0	S1	P0	P1
25	0–2	46	3 (0.63)	10 (0.84)	9 (0.83)	8 (0.85)	8 (0.87)	9 (0.90)	9 (0.90)
25	3–10	95	0 (0.38)	6 (0.79)	5 (0.77)	6 (0.83)	6 (0.85)	8 (0.88)	8 (0.89)
25	11–30	143	0 (0.67)	4 (0.75)	4 (0.75)	3 (0.77)	3 (0.79)	3 (0.86)	3 (0.86)
25	>30	30	0 (0.52)	0 (0.75)	0 (0.75)	0 (0.72)	0 (0.71)	0 (0.84)	0 (0.83)
100	0–2	46	11 (0.63)	23 (0.84)	24 (0.83)	26 (0.85)	26 (0.87)	25 (0.90)	25 (0.90)
100	3–10	95	0 (0.38)	26 (0.79)	27 (0.77)	28 (0.83)	29 (0.85)	32 (0.88)	33 (0.89)
100	11–30	143	0 (0.67)	8 (0.75)	10 (0.75)	9 (0.77)	10 (0.79)	8 (0.86)	8 (0.86)
100	>30	30	0 (0.52)	0 (0.75)	0 (0.75)	0 (0.72)	0 (0.71)	0 (0.84)	0 (0.83)
300	0–2	46	34 (0.63)	27 (0.84)	29 (0.83)	30 (0.85)	30 (0.87)	31 (0.90)	31 (0.90)
300	3–10	95	0 (0.38)	41 (0.79)	46 (0.77)	41 (0.83)	45 (0.85)	58 (0.88)	60 (0.89)
300	11–30	143	0 (0.67)	25 (0.75)	25 (0.75)	31 (0.77)	31 (0.79)	44 (0.86)	48 (0.86)
300	>30	30	0 (0.52)	0 (0.75)	0 (0.75)	3 (0.72)	3 (0.71)	5 (0.84)	6 (0.83)

Table 13: Top-100 overlap (registered tie order): number of variants shared by each pair of first-100 shortlists. Source: archived `out/metrics.json`, `top_k_overlap`.

	distance	kmer_assay	kmer_cons	S0	S1	P0	P1
distance	–	14	12	9	9	11	10
kmer_assay	14	–	67	60	59	66	65
kmer_cons	12	67	–	57	57	61	62
S0	9	60	57	–	96	88	87
S1	9	59	57	96	–	87	87
P0	11	66	61	88	87	–	98
P1	10	65	62	87	87	98	–

Table 14: Validation grid for the k -mer baselines. Average precision on the 25% group-held-out part of the training arm (4,376 variants, 282 groups, 134 disrupting; inner training 15,033 variants). Source: archived `runs/controls/receipt.json`.

Model	Learning rate	Leaves	Validation AP	Role
kmer_assay	0.03	15	0.2728	selected published
kmer_assay	0.03	31	0.2787	
kmer_assay	0.06	15	0.2920	
kmer_assay	0.06	31	0.2664	
kmer_assay	0.1	15	0.2869	
kmer_assay	0.1	31	0.2596	
kmer_cons	0.03	15	0.3124	published selected
kmer_cons	0.03	31	0.3035	
kmer_cons	0.06	15	0.3150	
kmer_cons	0.06	31	0.2933	
kmer_cons	0.1	15	0.3193	
kmer_cons	0.1	31	0.2970	

Table 15: Share of bootstrap draws in which the P@k difference was exactly zero, a measure of how discrete the P@k difference is at each budget. Source: `share_of_draws_exactly_zero` in the archived `paired_contrasts`.

Candidate	Against <code>kmer_cons</code>			Against <code>kmer_cons_selected</code>
	P@25	P@100	P@300	P@100
S0	13.60%	7.85%	4.20%	8.05%
S1	13.60%	6.15%	2.30%	7.30%
P0	11.25%	6.25%	0.00%	9.15%
P1	11.25%	4.55%	0.00%	6.70%

Table 16: Bootstrap design per run. Realised review depth across draws (each draw keeps the fraction k/N), draws, resampled groups and single-class draws skipped. Source: `realised_depth` in the archived `paired_contrasts`.

Nominal budget	Depth range	Mean depth	Draws	Groups	Single-class skipped
25	21–31	25.0	2,000	460	0
100	83–126	99.9	2,000	460	0
300	250–378	299.6	2,000	460	0
100 (selected baseline)	83–126	99.9	2,000	460	0
20 (tutorial subset)	15–26	20.0	200	58	0

Table 17: Tutorial subset (pipeline check only, not an evaluation result). Every eighth held-out group: 977 variants, 44 disrupting, 58 groups, budget 20, 200 draws. Shown only so that the archived `out-tutorial/` outputs are fully accounted for; these intervals should not be interpreted. Source: archived `out-tutorial/metrics.json`.

Candidate	P@20	AP	AUROC
S0	+0.050 [−0.095, +0.223]	+0.077 [−0.010, +0.173]	+0.052 [−0.011, +0.107]
S1	+0.100 [−0.053, +0.235]	+0.080 [−0.010, +0.177]	+0.062 [−0.035, +0.135]
P0	+0.100 [+0.000, +0.319]	+0.088 [+0.029, +0.156]	+0.084 [+0.021, +0.165]
P1	+0.100 [+0.000, +0.319]	+0.117 [+0.052, +0.183]	+0.118 [+0.051, +0.173]

E Selection flowchart

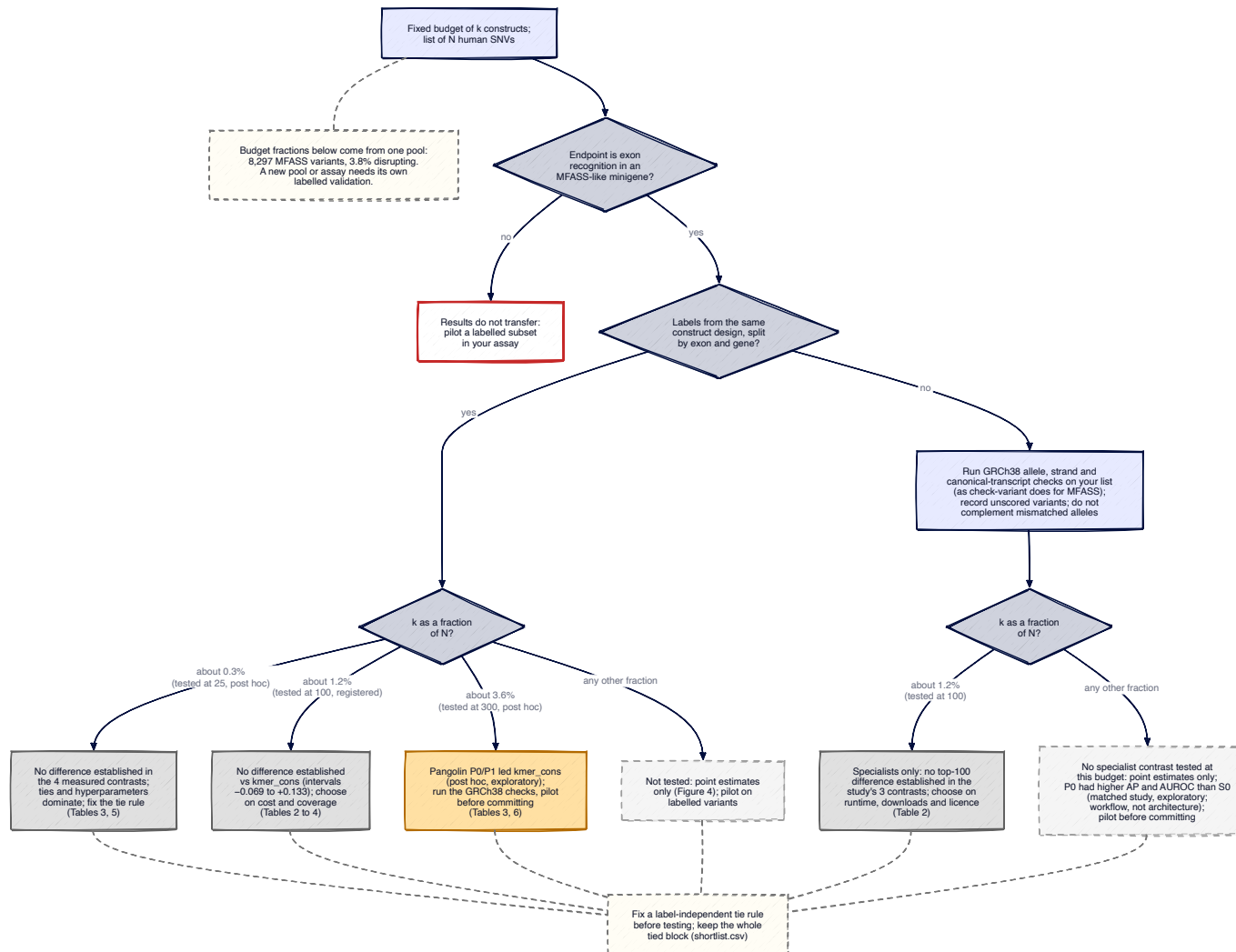


Figure 6: Selection flowchart (original Figure 6, converted from `article/assets/06-selection-flowchart.svg`; D2 source `article/assets/06-selection-flowchart.d2`). Questions: is the endpoint MFASS-like minigene exon recognition; do same-design labels exist; what fraction of the list is the budget. Terminals cite the original Tables 2–6 (here Tables 2, 3, 4, 6 and 7); untested fractions end at “not tested”, not at a recommendation. The vector figure can be zoomed; Table 9 gives the same routes in readable form.

F Provenance, supplement and evidence mapping

Supplement S1 (original article). The full original article, *Choosing a Splicing Score for a Fixed Minigene Budget: Baseline, SpliceAI and Pangolin on MFASS* (published 30 September 2026), is retained unchanged in the repository as `article/published-original.md` (byte-identical to the published source) and is embedded in this PDF as the attachment `S1-published-original.md`. The repository copy `article/original.md` differs only in local links. The original contains a frequently-asked-questions block and the reader-facing narrative that this manuscript restructures; every scientific result it reports appears in this manuscript, and the section-by-section mapping is in `evidence/paper-migration/content-coverage.md`.

Evidence ledger. `evidence/paper-migration/claims-ledger.json` links each substantive claim to the historical artifact (archive member or article text), a pointer inside it and its SHA-256. The archived outputs (`downloads/splice-shortlist-outputs.zip`) were read in memory without extraction; their member digests are in `evidence/reference-output-members.json`. The table generator (`scripts/paper_extract.py`) stops the build if any generated value that also appears in the original article differs from it (evidence-map items T2–T6, C-POP, C-OVERLAP, F4, M-STUDY); the recorded cross-check found no difference. Values that the original did not print are formatted directly from the digest-checked archive members.

Citation note. The companion README cites the MFASS paper as *Mol Cell* 2018 (its online date); this manuscript follows the original article’s issue citation (2019;73(1):183–194.e8) [4]. Literature claims were verified for the original article on 30 September 2026 and were not re-retrieved during migration.